data conference
SWITZERLAND

Extending Fabric monitoring for custom requirements

Marc Lelijveld – Data Platform MVP, Solution Architect @ Macaw

Iqbal Khan – Senior Data Analyst, Microsoft

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Presenters

Marc Lelijveld

Technical Evangelist & Data Platform MVP

- Joined Macaw in 2018
- Accidental Fabric Administrator at various enterprise clients
- Previously only focused on Power BI, till Fabric came along!
- Based in a tiny town next to Gouda - Netherlands

Presenters

Iqbal Khan

Admin Monitoring + OneLake Catalog – Reporting Lead @Microsoft

- Joined Microsoft in 2021 as a Power BI architect
- Moved to Fabric engineering in 2023
- Based in NYC with my wife Kellie and cat Jax
- Currently pursuing Masters of Analytics at UW-M

Agenda - Extending Fabric monitoring for custom requirements

1. Why monitoring? Which monitoring solution is right for me?
2. OneLake Catalog + Admin monitoring workspace deep-dive
3. Custom monitoring through Semantic Link
4. Session closeout, Q/A

Why monitoring? Monitoring can help with:

- Best practice and policy enforcement
- Audit and compliance
- Efficiency and reusability
- Cost optimization

Why monitoring?

As organizations continue to expand their use of Fabric, it is critical to have a governance strategy in place to ensure secure and effective use of your data.

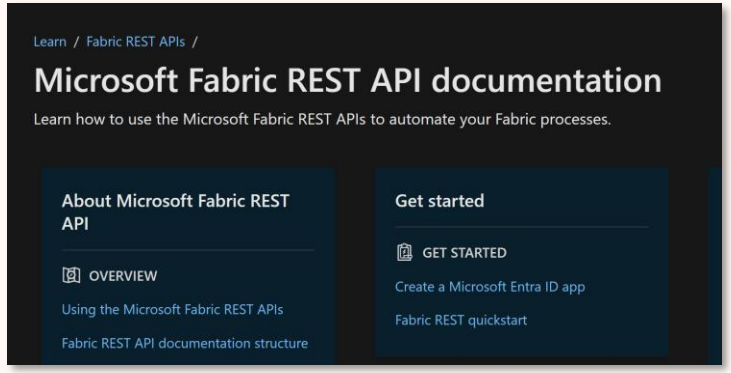
Monitoring can help with:

- Understanding data solutions needed to support business
- Ensuring safe and secure use of data
- Maximizing Fabric return on investment
- Reducing data outages
- Enforcing best practices and organizational policies

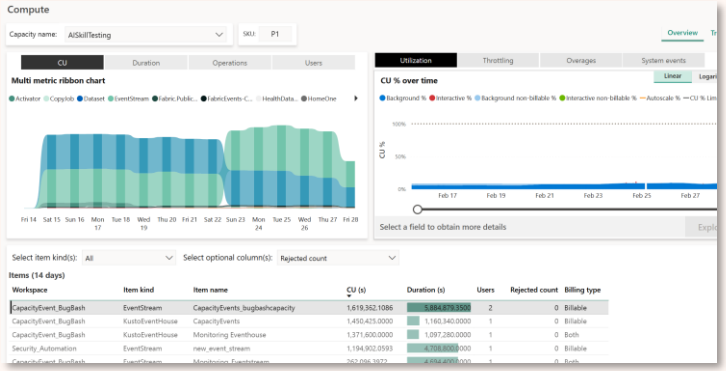
...and much more!

Which Microsoft monitoring solution is right for me?

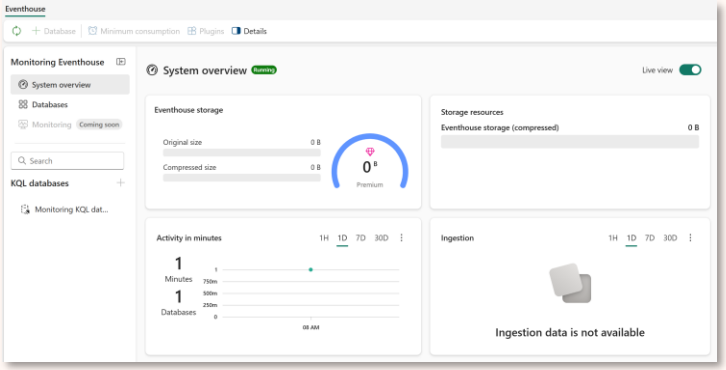
Admin monitoring – Out of the box reports on key governance scenarios | **Admin**



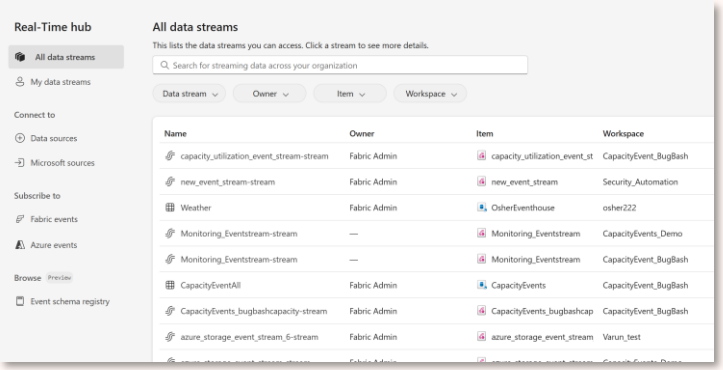
APIs – Flexible, programmatic data extraction & automation | **Admin\Developer**



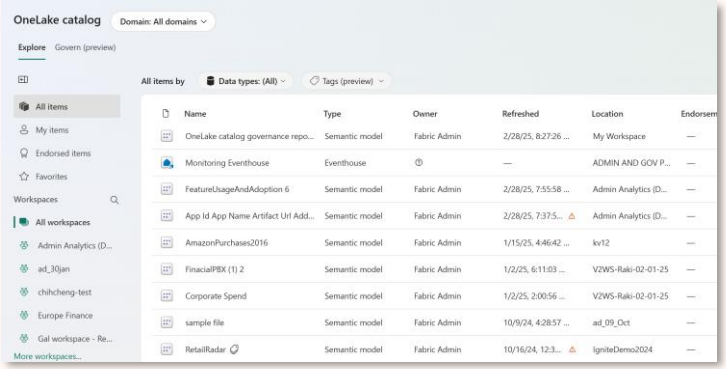
Capacity Metrics App – Capacity management and resource planning | **Tenant\Capacity Admin**



Workspace Monitoring – Diagnostic logging for Fabric items | **Developer**



Real Time Hub – Event streaming for Fabric workloads for responsive actions | **Developer**



OneLake catalog – Actionable lens into your data estate | **Developer**

OneLake Catalog – For creators

Public preview -> Demo

Central hub to find, explore, and manage the Fabric items you own

Overview (Explore tab)

OneDrive-like UI to browse and explore **all your Fabric items** in one place without losing context

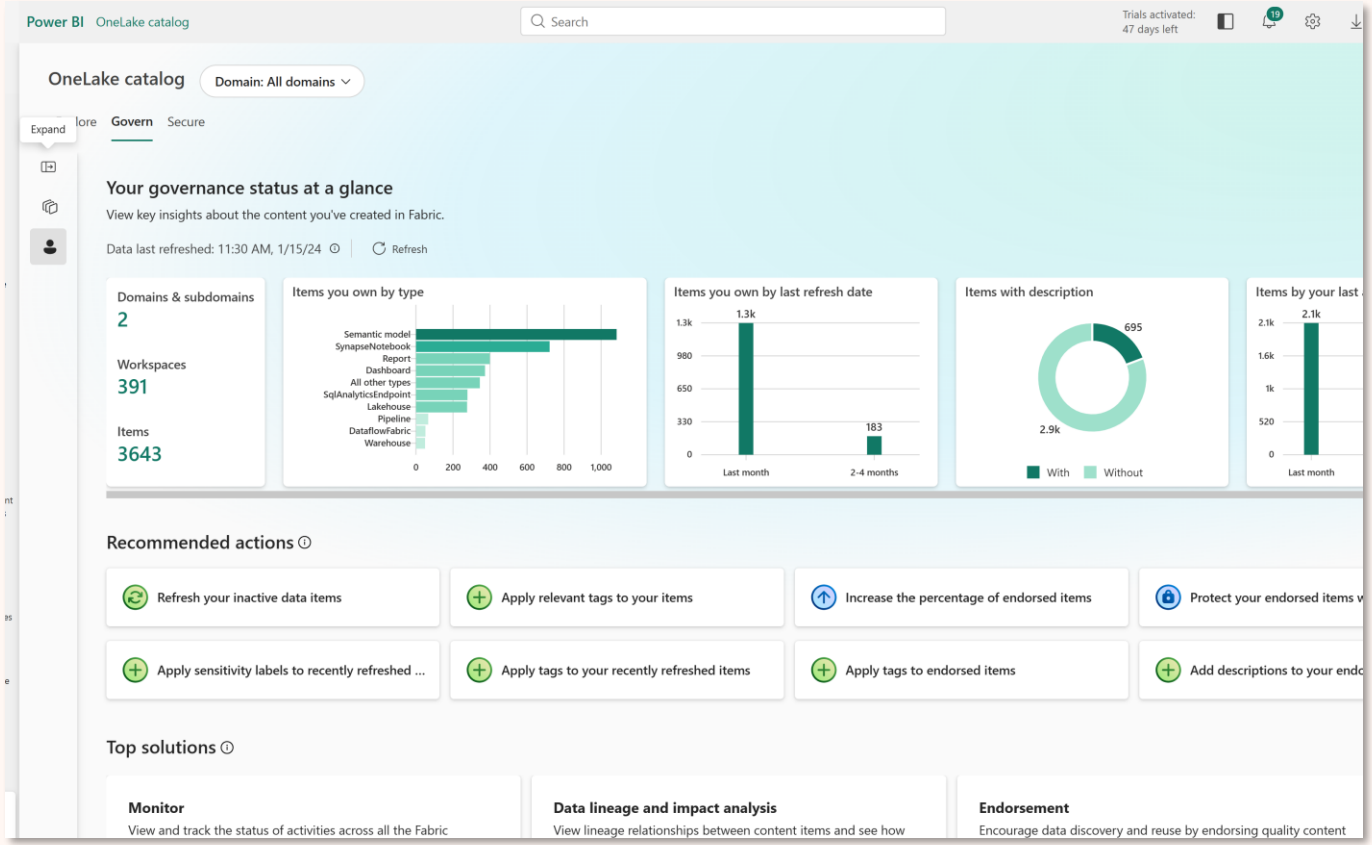
Item Details (Explore tab contd.)

Dive further into item details, highlighting last refresh time/status, permissions, external shares, lineage, and more

Act (Govern tab)

Dynamic actions to better secure and govern your data, plus built-in insights to visualize your data estate

**Not covered: Secure tab -> Managing permissions



OneLake Catalog – For admins

Coming soon -> Demo

Central hub to set up, manage, and protect the Fabric items <u>across your entire tenant</u>

Overview (Explore tab)

OneDrive-like UI to browse and explore **your entire data estate** in one place without losing context

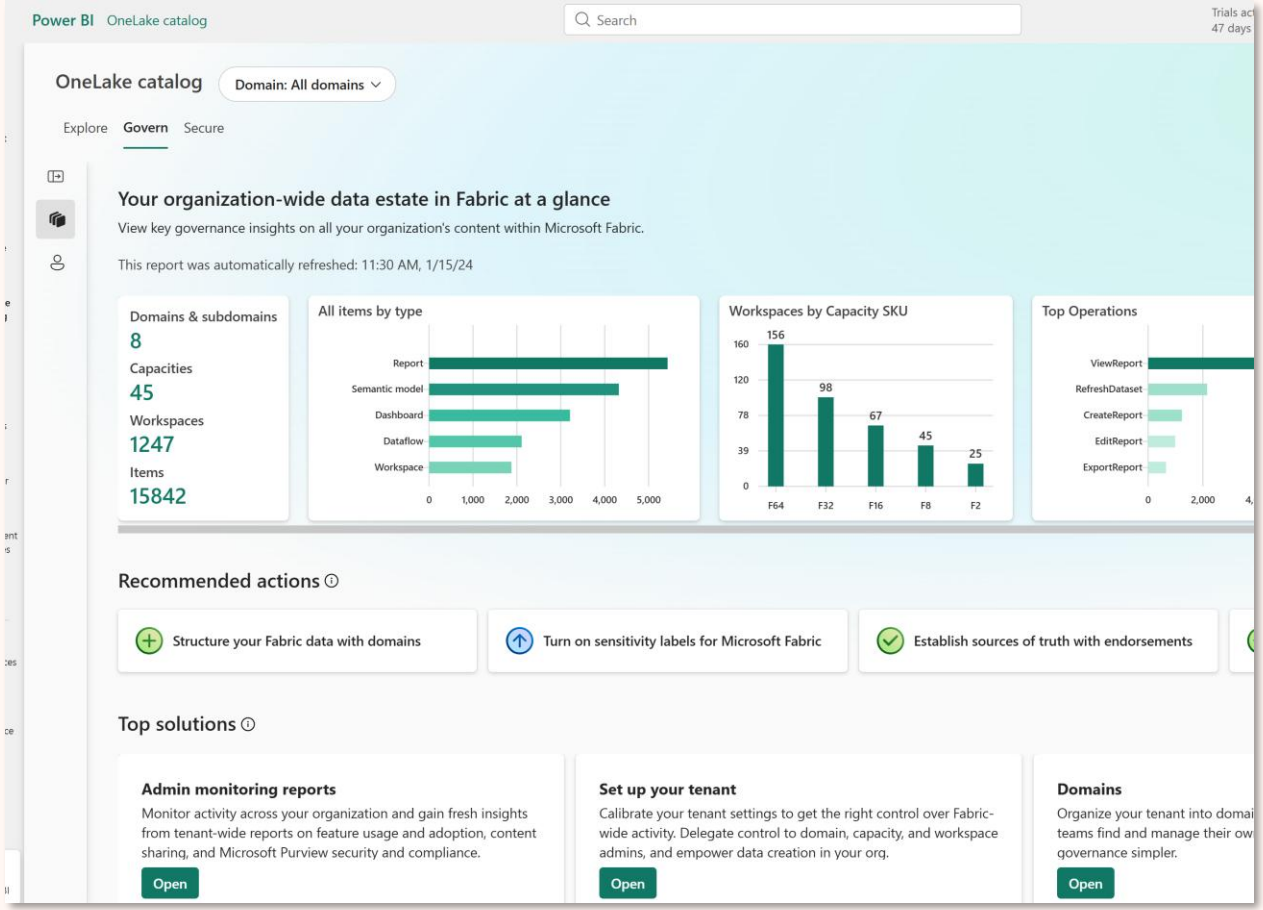
Item Details (Explore tab contd.)

Dive further into item details, highlighting last refresh time/status, permissions, external shares, lineage, and more

Take action (Govern tab)

Dynamic actions to better secure and govern **your entire data estate**, plus built-in insights to support governance efforts

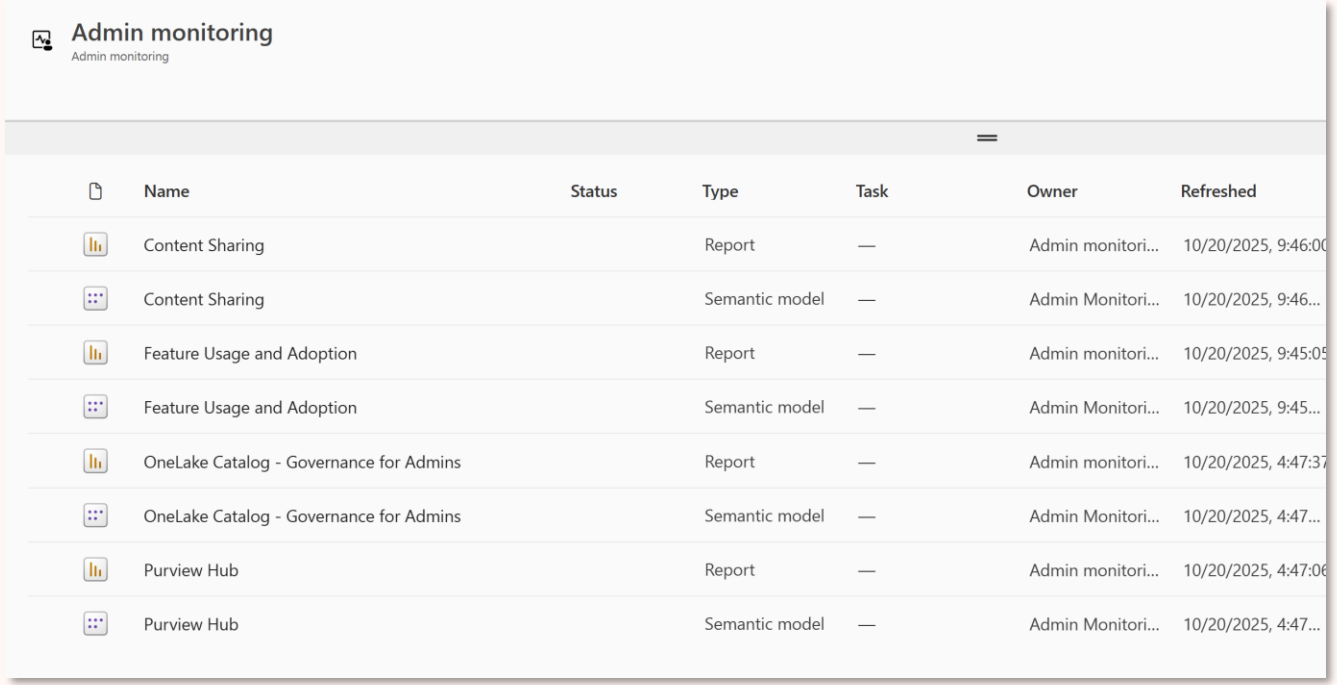
**Not covered: Secure tab -> Managing permissions



What is the Admin monitoring workspace?

Public preview -> Demo

- **One-stop shop** for enterprise reporting and analytics
- Includes **out of the box reporting** focused on Fabric tenant management scenarios
- Also includes **semantic models for customization**
- Available to **all tenants** regardless of licensing type or total # of users



Admin monitoring
Admin monitoring

Name	Status	Type	Task	Owner	Refreshed
Content Sharing		Report	—	Admin monitori...	10/20/2025, 9:46:00
Content Sharing		Semantic model	—	Admin Monitori...	10/20/2025, 9:46...
Feature Usage and Adoption		Report	—	Admin monitori...	10/20/2025, 9:45:05
Feature Usage and Adoption		Semantic model	—	Admin Monitori...	10/20/2025, 9:45...
OneLake Catalog - Governance for Admins		Report	—	Admin monitori...	10/20/2025, 4:47:37
OneLake Catalog - Governance for Admins		Semantic model	—	Admin Monitori...	10/20/2025, 4:47...
Purview Hub		Report	—	Admin monitori...	10/20/2025, 4:47:06
Purview Hub		Semantic model	—	Admin Monitori...	10/20/2025, 4:47...

Perfect – yet imperfect…
There are always custom requirements!

What Microsoft provides

- Platform structure and governance, like capacities, domains and workspaces
- Item metadata, describing platform artifacts
- Audit logs, showing facts about usage and adoption of the platform

What needs customization

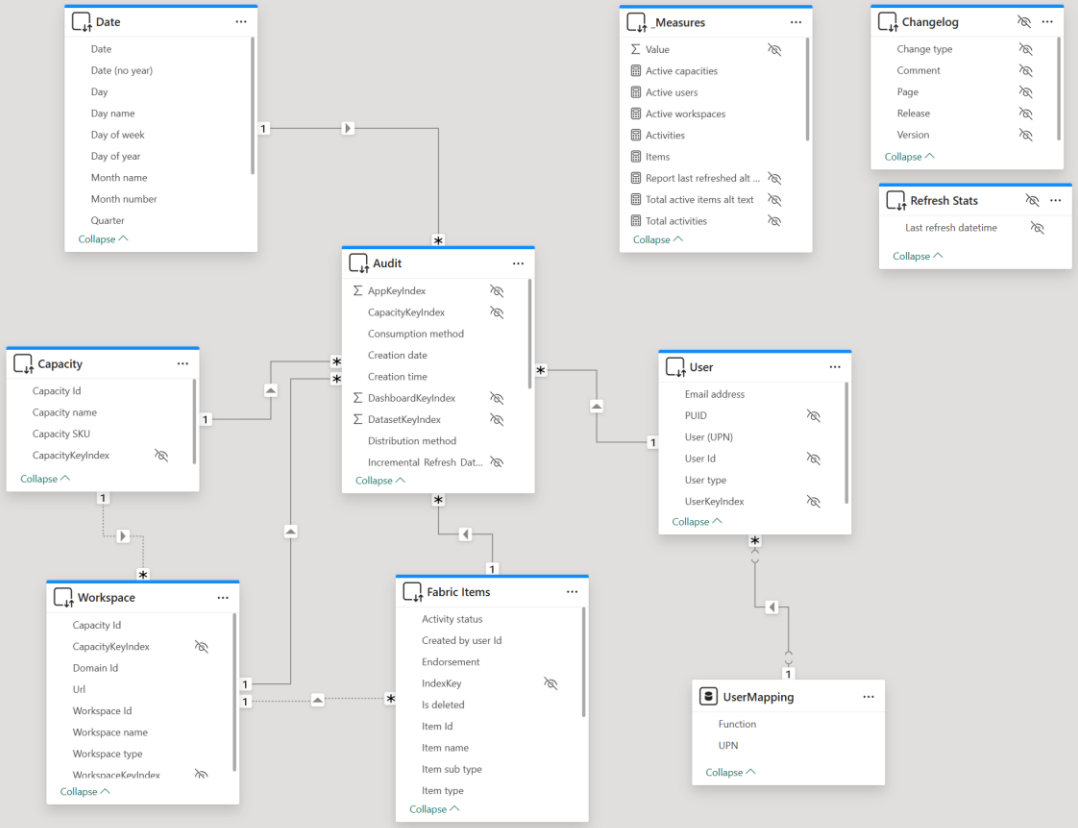
- Organizational structure
- Licensing (part of M365 admin)
- Cost monitoring components (capacity costs)
- Extensions to monitor cross-platform adoption

Enriching with your own data
Composite models

What are composite models:

- Connect to Power BI semantic models and Azure Analysis Services models using DirectQuery to build custom models using multiple sources
- In composite models, you combine 2 or more ***source groups*** into a single semantic model

Notice two different source groups → Admin monitoring data + custom org hierarchy



Extending Admin monitoring

Goals

- Easily combine data like org. hierarchy, licenses, etc.
- Trend analysis
- Long term storage of data

Non-goals

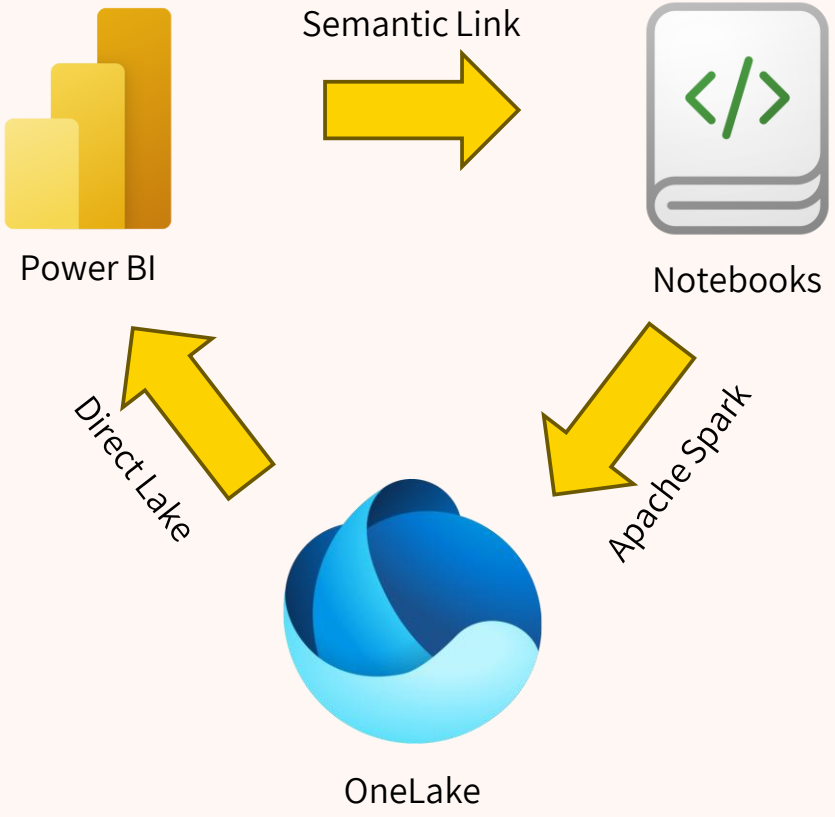
- Exporting to unmanaged files and carriers (like Excel)
- Distributing data across the organization

Introduction to Semantic Link

Semantic Link is a Microsoft Fabric feature that allows you to connect to Power BI Semantic Models from Fabric Notebooks.

This feature **only** exists and works in Fabric.

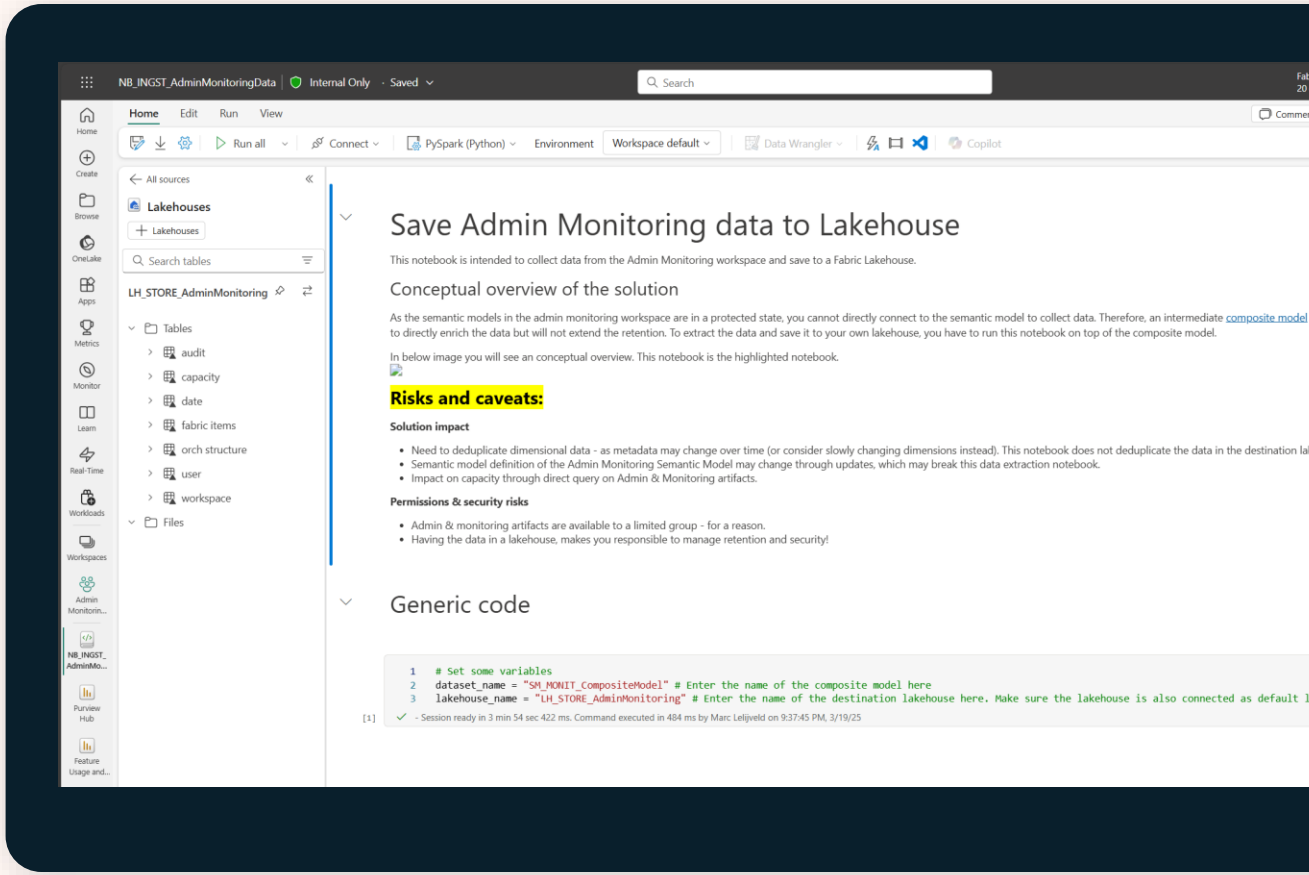
- Write DAX, Python, SQL, and informative queries directly against your semantic models
- Data as well as metadata
- Operate and automate tasks like refreshes, partitioning, creation of items, etc.
- More than just semantic models, enables automation of Fabric tasks in general
- Open-source extension Semantic Link Labs to do even more!



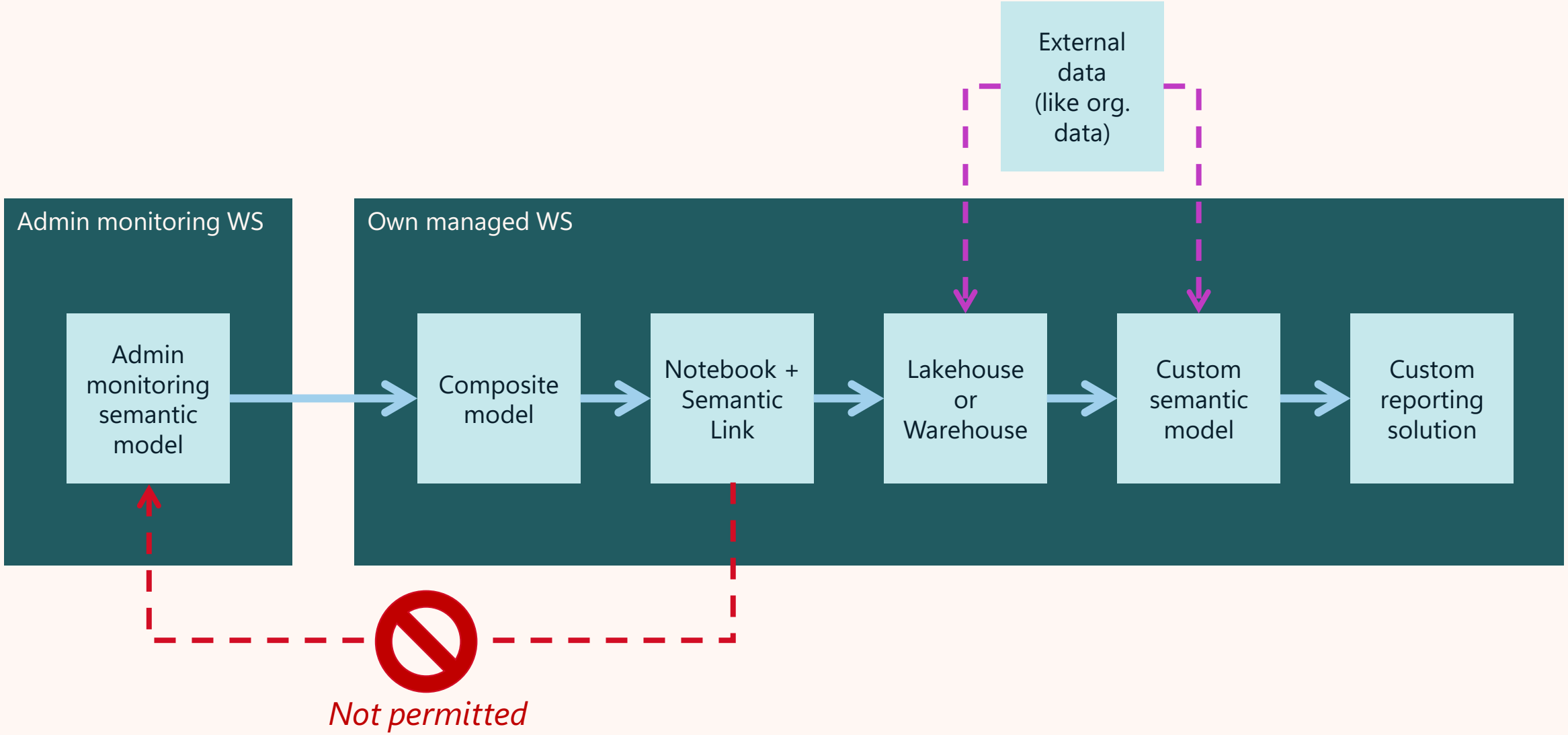
Quick look at Notebooks

- Code first
- Web-based interface
- Cell-based code blocks
- Runs on nodes (part of Fabric capacity)
- Often used languages incl. Python, Spark & Markdown

- Used by data engineers for data ingestion, prep, and transformations
- Used by data scientists for experiments and models



Solution overview (conceptual)



Risks and caveats

Solution impact

- Need to **deduplicate** dimensional data – as metadata may change over time (or consider slowly changing dimensions instead)
- Semantic model definition may **change** through updates, which may **break** your data extraction notebook
- Impact on **capacity** through DirectQuery on Admin monitoring semantic models

Permissions & security risks

- Admin monitoring artifacts are available to a **limited group** – for a reason
- Having the data in a lakehouse makes you responsible to manage **retention** and **security**!

Sharing your admin insights?

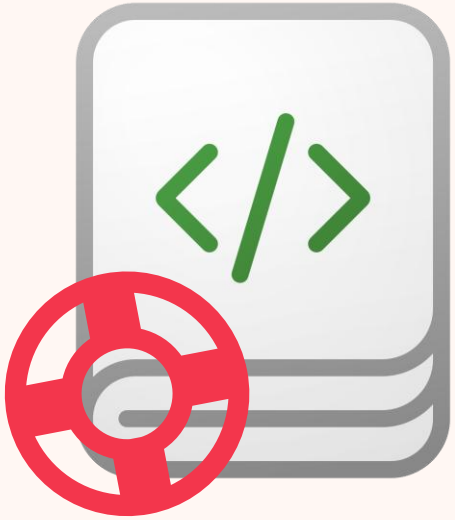
Sharing is caring!

- Every domain / workspace admin (should) want(s) insights on their own content
- Only share what is relevant to them, with customized security on top (RLS / OLS)

But…

- After every update, your security config might be overwritten
- Lot of manual effort to reconfigure every time

Link: https://data-marc.com/2025/10/14/automating-power-bi-semantic-model-security-with-fabric-semantic-link/



Semantic Link to the rescue!

Alternative approach:
Fabric Unified Admin Monitoring (FUAM)

Make sure to attend the FUAM session as well →

Oliver Engels Gellert Gintli Kevin Thomas

X-Ray your Microsoft Fabric Landscape with FUAM

One important bit of the Microsoft Fabric adoption roadmap is System oversight. In this session, we will have a look into what is possible to Monitor and Audit your Fabric Landscape. We will look into the out of the box functionality, what is on the roadmap and we will deep-dive in more advanced functionality by using the Fabric Unified Admin Monitoring Framework (FUAM). This comprehensive framework will include data from different APIs, Capacity Management App, Best Practise - and Vertipaq Analyzer to build an outstanding monitoring solution. We will show live monitoring possibilities with FUAM and our experience by using it in large Fabric landscapes.

ROOM 3 Wed 4:30 pm - 5:15 pm

Additional links and resources

Official public documentation:

- Admin Monitoring Workspace
  https://learn.microsoft.com/en-us/fabric/admin/monitoring-workspace
- Feature Usage and Adoption Report
  https://learn.microsoft.com/en-us/fabric/admin/feature-usage-adoption
- OneLake Catalog – Overview
  https://learn.microsoft.com/en-us/fabric/governance/onelake-catalog-overview
- OneLake Catalog – Govern your data
  https://learn.microsoft.com/en-us/fabric/governance/onelake-catalog-govern

Additional links and resources

Open-source solutions and accelerators:

- Marc's GitHub with scripts for extending Fabric monitoring
  https://github.com/marclelijveld/Fabric-Automation/
- Fabric Toolbox
  https://github.com/microsoft/fabric-toolbox/tree/main
- Fabric Unified Admin Monitoring (FUAM)
  https://github.com/microsoft/fabric-toolbox/tree/main/monitoring/fabric-unified-admin-monitoring
- Fabric Cost Analysis
  https://github.com/microsoft/fabric-toolbox/blob/main/monitoring/fabric-cost-analysis

Additional links and resources

Conference feedback

Session feedback

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